CIRC HQ STUDENT HANDOUT | GRADE 5

Rube Machine

Goal: Build a reliable chain reaction in which at least three distinct steps complete one simple task.

35 minutes | 7 steps | Engineering and Design | Playbook B meeting 20, grade 5 path

Ask | Plan | Build | Test | Improve

You are done when

Demonstrate two complete runs and explain the handoff that required the most revision.

You will use

- Dominoes
- Cardboard ramps
- Cups
- String
- Craft sticks
- Balls
- Tape
- Final-task cards

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Get ready — 3 min

- Choose one final task card.
- Place moving parts inside the table boundary.

2 Step 2: Motion safety — 3 min

- Use only rolling tabletop parts.
- Keep hands outside a moving path.
- Stop any part that leaves the table.

3 Step 3: Plan backward — 5 min

- Draw the final task first.
- Add three earlier steps.
- Mark where each part starts the next part.

4 Step 4: Test each step — 8 min

- Build one step.
- Test that step one time.
- Fix the first part that does not work.

5 Step 5: Connect the chain — 8 min

- Join the tested steps.
- Try one full run.
- Fix the first place where motion stops.
- Try one more full run if time remains.

6 Step 6: Reset the parts — 5 min

- Stop every moving part.
- Sort parts into labeled bins.
- Remove tape scraps.

7 Step 7: Exit check — 3 min

- Circle the hardest move from one part to the next.
- Write the fix that worked.

What to turn in

Two complete runs and the handoff you fixed.

No computer? Another way to do it

Plan the chain on paper and act it out.

Finished early? Next challenge

One More Link: Add one useful step without reducing the machine's two-run reliability.

Stay safe

Keep moving parts inside the table boundary and use no flame, sharp projectile, or powered launcher.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.